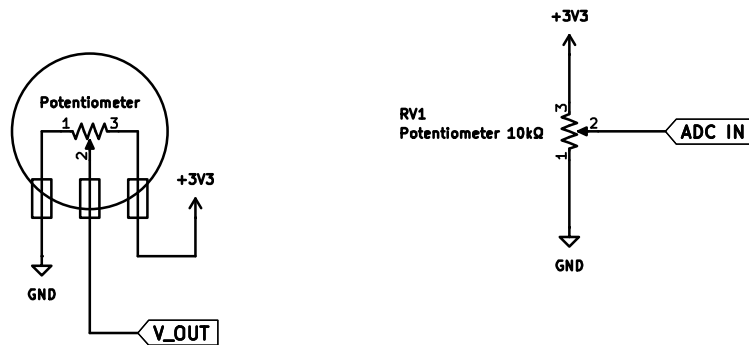
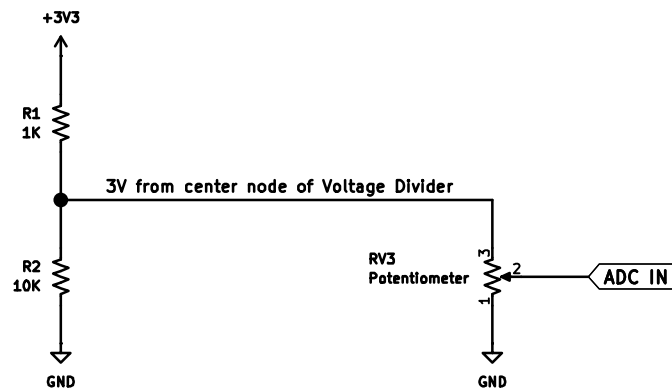


POTENTIOMETER



- Potentiometers or "Pots" allow us to manually vary its voltage output.
- Essentially, it's a Voltage Divider that outputs scaled voltage within the range of voltages applied to pins 1 and 3.
- In the above example, pin 2 will output between 0V [pin 1] and 3.3V [pin 3] depending on the pot's position.
- Use 10kΩ Potentiometers on Microcontrollers for best results.

POTENTIOMETER – Adjusted Voltage Range



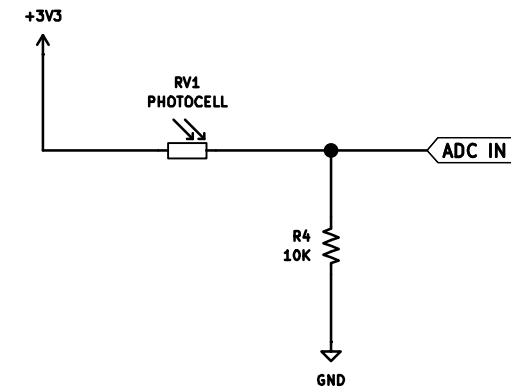
- In this circuit, maximum voltage output of pot is reduced to 3.0 Volts by means of a voltage divider.
- This avoids the nonlinear range [3.0V to 3.3V] on the ESP32-S3 ADC.

Light-Dependent Resistor (LDR) / PHOTOCELL

A PhotoCell is a Light-Dependent Resistor (LDR).

We can use it in a Voltage Divider Circuit, as shown below.

- When light levels increase...
- PhotoCell resistance decreases.
 - Voltage output increases.
 - ADC read values increase.



Sheet: /
File: ADC.kicad_sch

Title: ADC: Potentiometer & PhotoCell

Size: A4 Date:

KiCad E.D.A. 10.0.1

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